Numericals. 4. leaching

Page No. Date

Q. Write material balance of single stage leaching.

Ans: **single Stage leaching:**

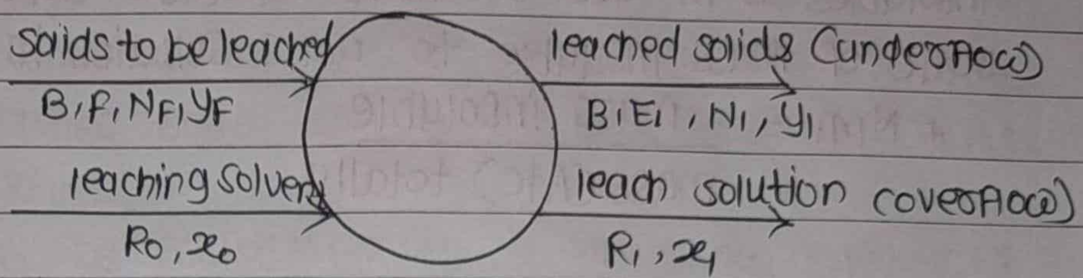

Fig. single Stage leaching.

1) Consider the single stage leaching operation in which the solid to be leached are in contacted with leaching solvent and then the resulting insoluble Phases are physically separated and the effluent streams are withdrawn as leached solvent and leach soln.
2) The circle in figure represents a stage in which the entire operation of leaching.
3) since in many cases clear liquid leach solution is obtained
4) for no solid in the overflow and solubility of B in A:
mass of insolubles (B) in leached solid = mass of insolubles in the solids to be leached.
5) material Balance of B:
$$B = N_F F = E_1 N_1$$
6) material balance of solute (C):
C in solids to be leached + C in leaching solvent = C in leached solid + C in leach soln.
$$\therefore F y_F + R_0 x_0 = E_1 y_1 + R_1 x_1$$
7) material Balance of solvent (A):
$$\therefore F(1-y_F) + R_0(1-x_0) = E_1(1-y_1) + R_1(1-x_1)$$
8) material Balance of solution (A+C):
$$F + R_0 = E_1 + R_1 = M_1$$

9) mixing of the solids to be leached and leaching solvent produces a B-free mixture M_1, so that the conc^n of insolubles B in the resulting mixture expressed as the ratio of mass insoluble to mass solution in N_{M_1}.

$$\therefore N_{M_1} = \frac{\text{mass insoluble}}{\text{mass }(A+C)\text{ totally}}$$

$$= \frac{B}{F+R_0} = \frac{B}{M_1}$$

10) The conc^n of solute C in resulting mixture expressed as the weight fraction of C on a B-free basis in the resulting mixture, y_{M_1} is determined by the material balance of C.

$$\therefore F y_F + R_0 x_0 = E_1 Y_1 + R_1 x_1 = y_{M_1} M_1$$

$$\therefore F y_F + R_0 x_0 = y_{M_1} M_1$$

$$\therefore y_{M_1} = \frac{F y_F + R_0 x_0}{M_1}$$

but $M_1 = F + R_0$

$$\boxed{\therefore y_{M_1} = \frac{F y_F + R_0 x_0}{F + R_0}}$$

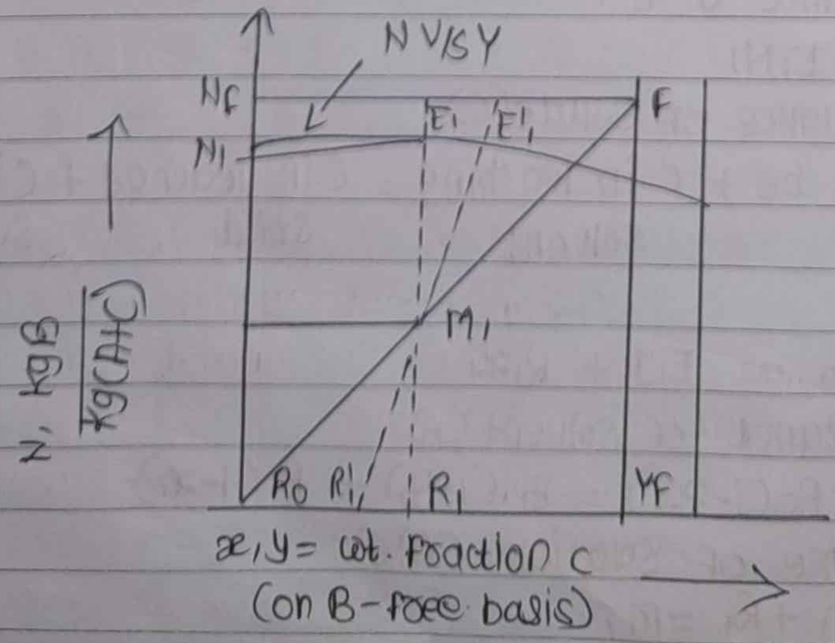

Fig. single stage leaching.

Q. Give the material Balance for the continuous crosscurrent leaching assuming variable underflow and no insoluble in the overflow.

Soln:
1) This is simply an extension of single stage leaching in which leached solids are sucessively contacted with fresh leached solvent.
2) This may be performed in a batch or continuous fashion.
3) Here the leached solid from any stage act as a feed solids to the next so more and more amount of the solute is removed from the solids as we move along the cascade.
4) Below fig shows a three-stage crosscurrent leaching operation in which unequal amounts of the leaching solvent of the same composition are used in all three stages.

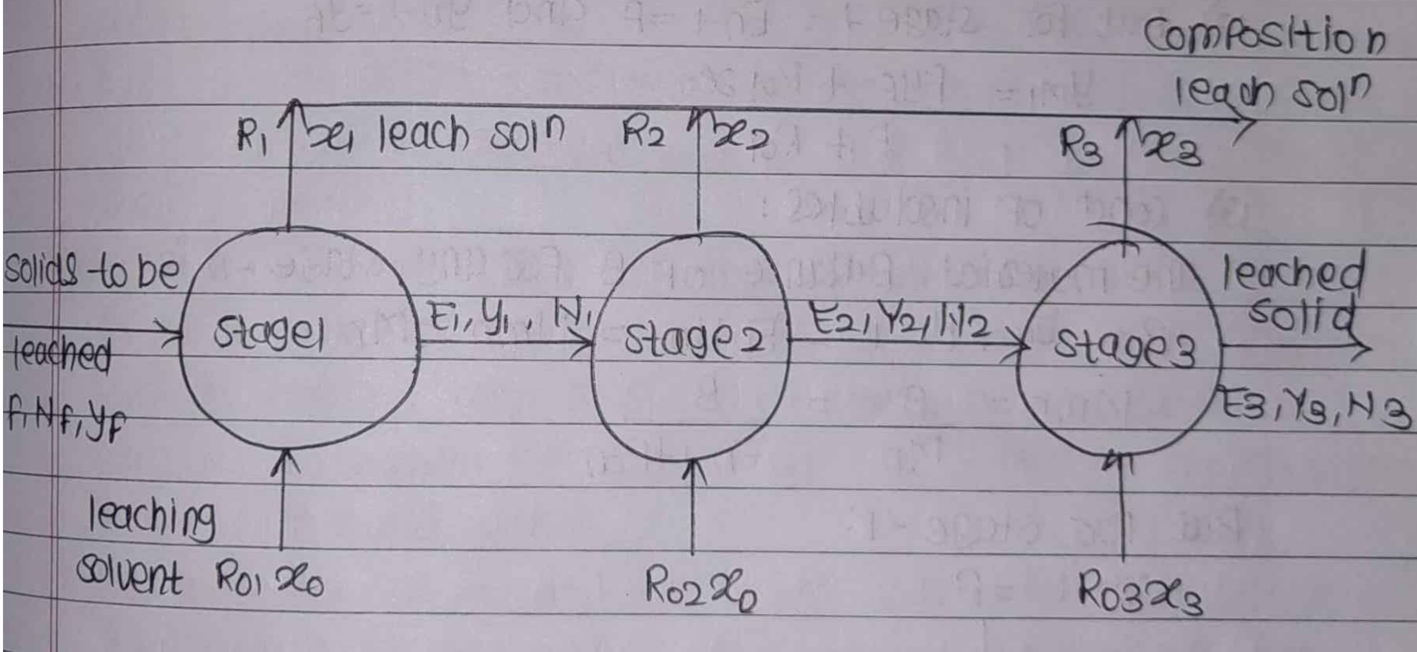

Fig. Three-stage crosscurrent leaching.

5) Assume that there are no insolubles in the leach soln so that B in solids to be leached is equal to B in leached solids from all stages.
6) material balance of insolubles (B) for any stage-n:

$$B = E_{n-1} N_{n-1} = E_n N_n$$

7) For stage-1 it becomes,

$$B = E_1 N_1 = N_F F$$

Page No.
Date

8) solution (A+C) balance for any stage-n

$$E_{n-1} + R_{on} = E_n + R_n = M_n$$

9) for stage 1 : $E_{n-1} = F$

$$\therefore F + R_{01} = E_1 + R_1 = M_1$$

10) for stage 2: It becomes

$$E_1 + R_{02} = E_2 + R_2 = M_2$$

11) Balance of solute (C) for any stage-n:

$$E_{n-1} y_{n-1} + R_{on} x_0 = E_n y_n + R_n x_n = M_n y_{mn}$$

$$\therefore y_{mn} = \frac{E_{n-1} y_{n-1} + R_{on} x_0}{M_n}$$

$$y_{mn} = \frac{E_{n-1} y_{n-1} + R_{on} x_0}{E_{n-1} + R_{on}}$$

12) But for stage-1 : $E_{n-1} = F$ and $y_{n-1} = y_F$

$$y_{m1} = \frac{F y_F + R_{01} x_0}{F + R_{01}}$$

13) concn of insolubles:

The material Balance of B for any stage-n is:

$$B = E_{n-1} N_{n-1} = E_n N_n = N_{mn} - M_n$$

$$\therefore M_{mn} = \frac{B}{M_n} = \frac{B}{E_{n-1} + R_{on}}$$

But for stage-1:

$$E_{n-1} = F$$

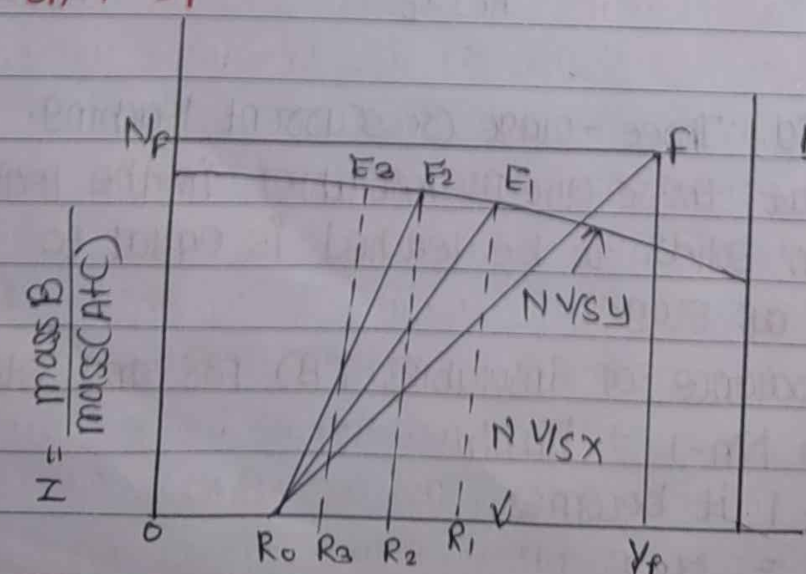

Fig. three stage crosss-current leaching.

x, y > wt. fraction C, B-free basis.

Page No.
Date

Q. Explain Bollman Extractor / Basket extractor.

Ans: **Bollman Extractor:**

1) It consists of a vapour tight vertical chamber in which a series of Perforated baskets are attached to chain conveyor.
2) The basket are Provided with Perforations at bottom.
3) The basket are loaded with the Flaky solids when each of the basket comes at the top of the downward side.
4) They are sprayed near the top of the chamber with the dilute solution of oil in solvent as they travel downward
5) As the basket travel downward the solids are leached with half miscella in Parallel Flow so that the solvent extracts more oil
6) The solution sprayed Percolates through the solids From basket to basket and collects at the bottom of the chamber in the right hand sump as the final strong soln of the oil and is removed out of the chamber from the outlet Provided
7) on the upward side of the extractor the Partially Extracted solids travel upwards and are leached Further countercurrently with a spray of Fresh solvent to obtain the dilute solution of the oil at the bottom of the chamber in the left hand sump.
8) on this side of travel the Fresh solvent is sprayed on the meal near the top which the Percolates through the meal from basket to basket.
9) on this side, meal moves upward and solution Flows downward and so the countercurrent flow is achieved.
10) when the basket are at top of their travel, they are inverted and Fully exhausted solid is discharged into the hopper from which it is removed by screw conveyor.
11) usually conveyor speed is one revolution per hour and solvent to Fresh meal ratio is one.

Page No.
Date

Q. Explain Pachuca tank:

Ans: Pachuca tank:

1) It is frequently used in the metallurgical industries.
2) It is operated in a batchwise fashion and thus provides a single stage leaching.
3) Here compressed air is used for agitation.
4) It consist of of a cylindrical tank with conical bottom.
5) It can be constructed out of metal, wood or concrete.
6) lead lining may also be done in certain cases depending upon the nature of the leaching liquid.
7) It is fitted ~~in a~~ with an arrangement for introducing compressed air.
8) It is provided with central tube that act as an air lift, the bubbles of air rising through the central tube cause the upward flow of liquid. ~~and~~
9) suspended solids in the tube that consequently result in continuous vertical circulation of the tank contents.

10) Air jets are provided at the conical portion of the tank for dislodging any material that settles out.

11) After leaching is complete, agitation is stopped and the solids are allowed to settle ~~out~~.

12) The clear supernant liquid is decanted from the top by withdrawing through discharge connections located at an appropriate height in the side of the tank.

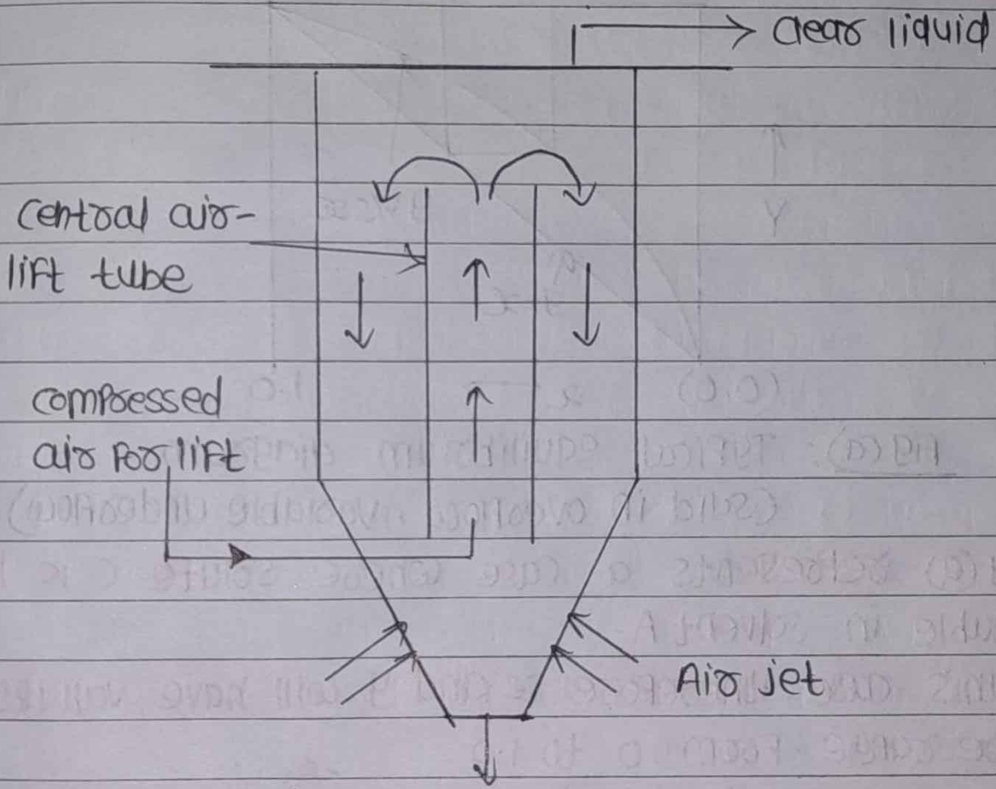

Fig. Pachuca Tank.

Q. Discuss on graphical representation of equilibrium characteristics of leaching operation with diagram and proper notations.

Ans: Graphical Representation of Equilibrium Conditions in leaching:

1) In case of leaching operation, when the equilibrium data are plotted on rectangular co-ordinates with N as a co-ordinate and x, y as abscissas. we get one of the types of the equilibrium curves

Page No.
Date

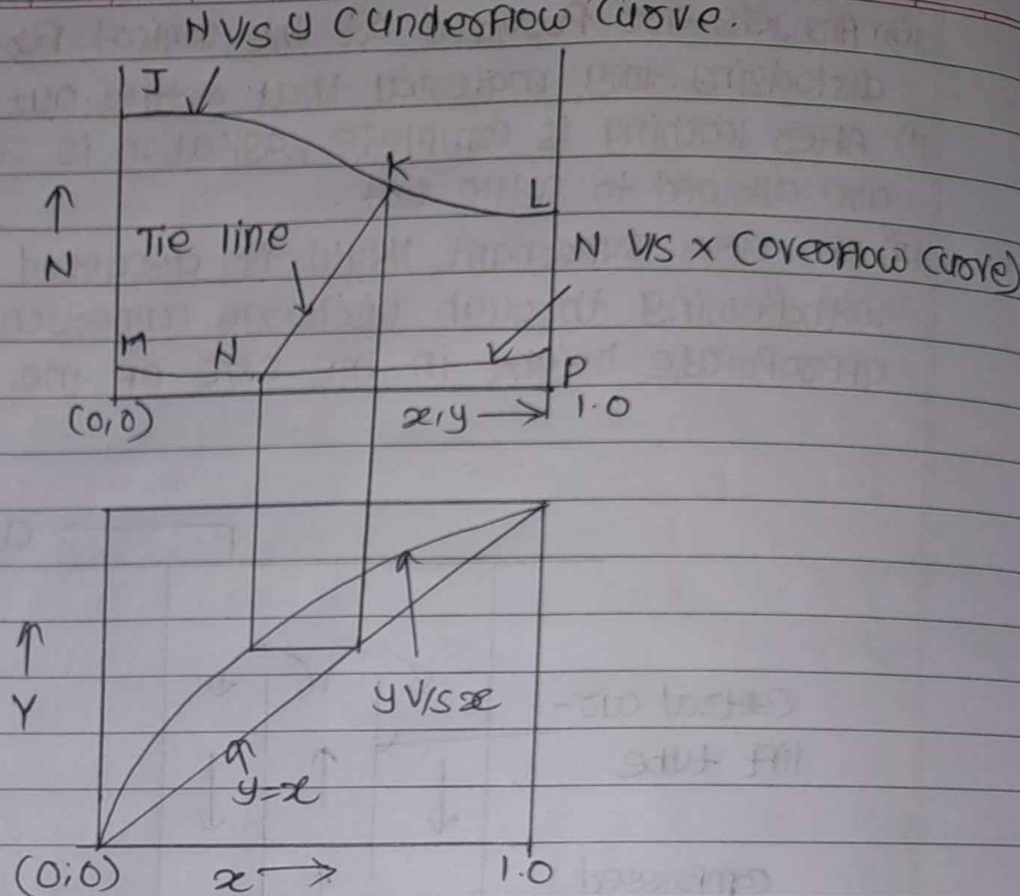

Fig (a): Typical equilibrium diagram
(solid in overflow, variable underflow)

1) Fig (a) represents a case where solute C is infinitely soluble in solvent A.
2) In this case, therefore x and y will have values over the entire range from 0 to 1.0
3) The curve JKL represents the seperated solid as expected in actual practice.
4) The curve JKL is a locus of underflow compositions.
5) The curve MNP represents the clear soln withdrawn and thus it is a locus of overflow composition.
6) Here the curve MNP lies above the N=0 axis which indicates the overflow contains same amount of insoluble solid B.
7) Here the tie lines are not vertical and this will happen.
(i) if enough time of contact with leaching solvent to dissolve all solute is not provided.

Page No.
Date

(ii) if adsorption of the solute occurs preferentially
(iii) if the solute C is soluble in the insoluble solid B

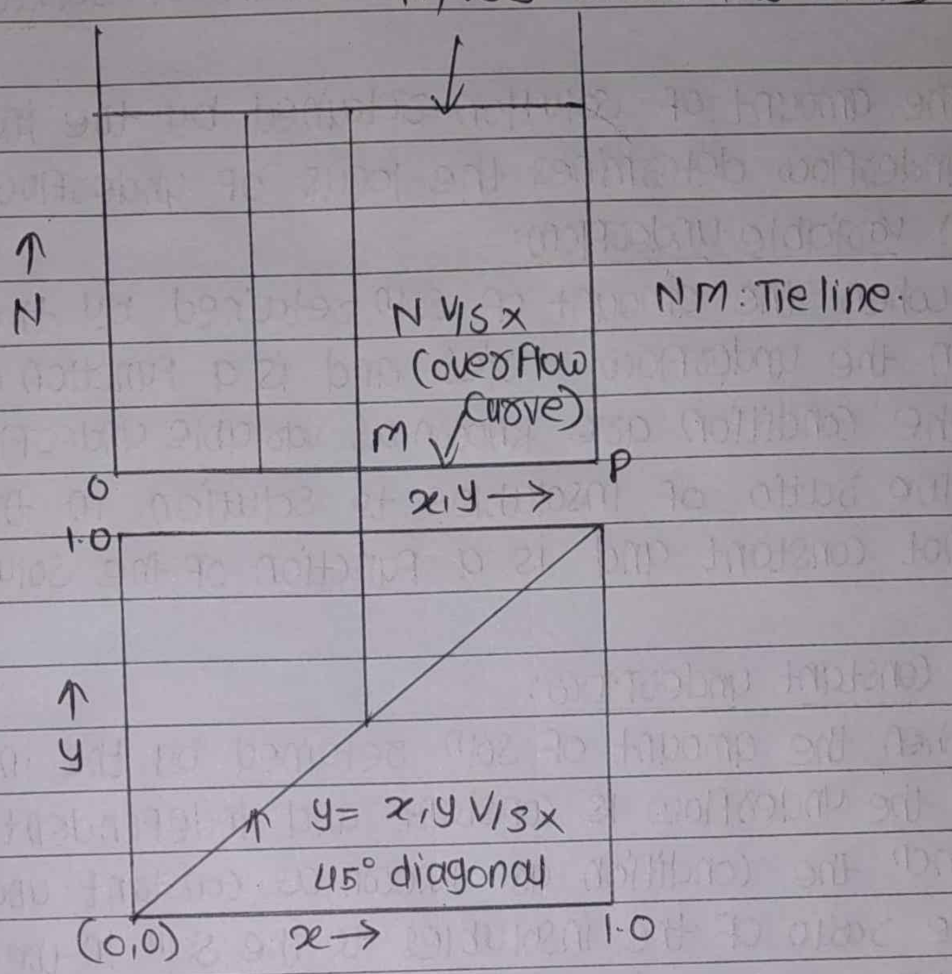

Fig (b). Typical Equilibrium diagram
(No solid in overflow, constant underflow)

(1) Fig (b) represents a case where the insoluble solid is non-adsorbent

2) It represents a chase where adsorption of solute does not occur so that clear soln is withdrawn and soln associated with the solid in the underflow have the same composition as the overflow and the tie lines are vertical and thus an x-y curve coincides with the 45° diagonal and a distribution coefficient, $m = y^*/x$ equals unity.

3) Here the overflow contains no insoluble solid B either dissolved or suspended so that the locus of overflow compositions is represented by the x-axis.

Page No.
Date

4) line JK representing underflow is horizontal which indicates that the solution retained by the insoluble solid B in underflow is constant and independent of solute concn.

⇒ The amount of solution retained by the insoluble solids in the underflow determines the locus of underflow compositions.

(1) variable underflow:

- when the amount of soln retained by the insoluble solid in the underflow varies and is a function of solute concn the condition are known as variable underflow.
- The ratio of insolubles to solution in the underflow is not constant and is a function of the solute concn value.

(2) constant underflow:

- when the amount of soln retained by the insoluble solid in the underflow is constant and independent of the solute concn the condition are known as constant underflow.
- The ratio of the insolubles to the soln in the underflow is constant for all the stages.

Q. Explain applications of leaching and factors affecting the rate of leaching.

Ans: Applications of leaching operation:

1) Preparation of tea and coffee
2) Extraction of perfumes from flowers
3) Extraction of sugar from sugar beets using hot water
4) Extraction of medicinal compounds
5) Extraction of tannin from tree barks using water.

Factors affecting rate of leaching:

(1) Particle Size:

- Smaller Particle Size Provide a greater interfacial area betn the solid and liquid which in turn results in a higher rate of transfer of material.
- But the Particle size should not be too small, otherwise the seperation of the solid from the liquid and drainage of solid residue will become more difficult.

(2) Solvent:

- The liquid used should be good selective solvent and should have a low viscosity in order to circulate it freely.

(3) Temperature:

- The solubility of the soluble solute material to be leached increases with temperature, so high temperature favours the leaching operation.
- Considering other factors there should be certain upper limit to the temperature.

(4) Agitation:

- Agitation increases eddy diffusion and thus increases the rate of transfer of material from the solid surface to the bulk solution.
- Agitation

Q. Basic steps in leaching operation.

Ans: 1) Contacting the solid with the selective solvent - the liquid in order to dissolve the soluble solute in the solvent

2) Seperation of insoluble Phases.

3) Seperation of extract from exhausted solid.

Page No.
Date

3) seperation of solvent from extract followed by Purification of Product
4) Recovery of solven from moist solid.